SLM Isolation Assessment

Sarnataro Federica M63001769
fed.sarnatero@studenti.unina.it

Tafuri Francesco M63001712
fran.tafuri@studenti.unina.it

***Abstract*—L'abstract deve introdurre il problema dell'esecuzione dei modelli in locale in ambito real time, spiegare dove gli llm possono essere utili nei sistemi embedded, con particolare riferimento ai compiti che prevedono prendere decisioni. Inoltre bisogna introdurre quali sono i problemi da affrontare nel deployment dei modelli su dispositivi embedded, con particolare attenzione sulla necessità di avere garanzie di tipo real time e perché sono necessari i meccanismi di isolamento. Spiegare l'approccio che abbiamo utilizzato, relativo al confronto tra due modelli basati su LLM, uno puro (Gemma) l'altro che funziona come decision model appartenente alla stessa categoria di jev.**

***Index Terms*—system one, system two**

I. Infrastruttura

L'obiettivo è quello di confrontare due modelli basati su due approcci differenti, definiti **system one e system two** in termini di: **responsività, stabilità, accuratezza e sensibilità alle interferenze**:

- **Gemma 4 31B** quantizzato ad 8 bit: si tratta di un LLM classico costituisce l'approccio *System two*
- **Semif OpenJev** basato su Qwen 3.5 4B non quantizzato: Si tratta di un LLM in cui è stato rimosso il layer autoencoder, al fine di leggere in output direttamente le probabilità dei logi, costituisce l'approcco *System One*

La differenza tra sistemi di tipo *System One* e *System Two*, proviene dalla psicologia, si tratta della classificazione effettuata da Daniel Kahneman nel suo saggio "Thinking fast and slow". L'utilizzo di questa classificazione nell'ambito dell'intelligenza artificiale lo si deve al team di *Typesafe AI* che nel descrivere le peculiarità del loro primo modello Jev, definisce gli LLM tradizionali di tipo *System Two* ovvero sistemi molto precisi, ma lenti nel fornire una risposta, in contrapposizione ai moodelli di tipo *System One*, cui Jev appartiene, molto rapidi nel prendere decisioni, ma meno precisi.

TABLE I
Differenze tra i modelli

	Gemma	**Semif**
input	Testo in linguaggio naturale non strutturato	Testo strutturato in JSON, con stato e tag tra cui scegliere
Output	Testo in linguaggio naturale	Probabilità del tag scelto e confidenza
Engine	llama.cpp su GPU	PyTorch su GPU
Modello di base	Gemma 4 31B, quantizzato a 8 bi	Qwen3.5 4B, non quantizzato (BF16)

La differenza sostanziale tra i due modelli, è che non disponendo del layer autoencoder, Semif a differenza di Gemma, non attraversa la fase di decode per fornire il risultato dell'inferenza, ma elabora in parallelo tutti i token dell'input e fornisce la risposta.

II. Hardware e software

La piattaforma su cui sono stati eseguiti gli esperimenti, è la Nvidia Jetson AGX Orin, equipaggiata con 64GB di RAM LPDDR5 condivisa tra CPU e GPU. Sulla board è installato il sistema operativo Ubuntu 22.04 CUDA con installato Jetpack 6.2.1. Il middleware utilizzato per la comunicazione tra sensori e modelli è ROS 2 Humble. Per l'esecuzione dei modelli, è stato usato pytorch per lanciare l'inferenza con Semif, mentre l'engine llama.cpp è stato usato per lanciare l'inferenza con Gemma. Infine per testare i modelli in presenza di workload paralleli è stata usata la suite stress-ng, per la generazione dei workload.

III. Scenario di test hard real time ed infrastruttura software

Ciascun modello prende decisioni per un robot a guida autonoma sulla base dei dati letti dai sensori: velocità, distanza dall'ostacolo e temperatura del motore. Le azioni possibili sono `proceed`, `slow_down` e `stop`. I modelli in oltre possono eseguire singolarmente o in contemporanea, operando sugli stessi dati.

L'ambiente di test simulato è costituito da 4 nodi ROS:

- *sensor_sim* è il nodo che simula il campionamento dei sensori e pubblica sul topic `/sensors`
- *llm_gen_node* si occupa di leggere i dati dal topic `/sensors` e di effettuarne il parsing per Gemma e pubblica il risultato dell'inferenza sul topic `/decisions/gemma`
- *semif_node* legge i dati dei sensori sul topic `/sensors` e ne effettua il parsing per semif e pubblica i risultati dell'inferenza sul topic `/decisions/semif`
- *logger_node* legge i dati dai topic legge i dati dai topic `/decisions/gemma` e `/decisions/semif` e scrive le statistiche in un file `.csv`

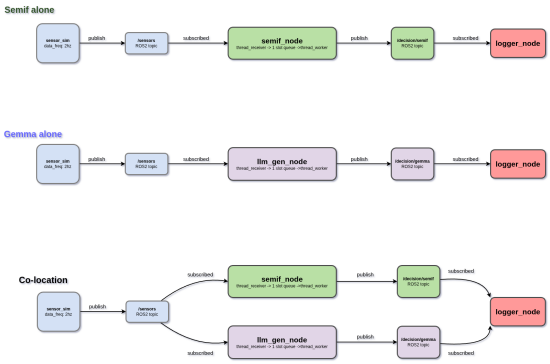


Fig. 1. schema nodi in diverse configurazioni.

A. Formato dei messaggi ROS

I messaggi pubblicati da *sensor_sim* sul suo topic hanno il seguente formato

```
{
  "seq": "data_sequence_number",
  "t_gen_ns": "generation_istant",
  "expected": "expected_action",
  "row": {
    "id": "data_sequence_number",
    "state": {
      "obstacle_distance_m": "random",
      "motor_temperature_c": "random",
      "speed_m_s": "random"
    },
    "question": "Which action should the robot
        take right now?",
    "options": [
      {"id": "proceed", "description": "Keep
          moving at the current speed: the
          obstacle is at least 1.5 m away and
          the motor temperature is at most 75
          degrees C."},
      {"id": "slow_down", "description": "
          Reduce speed: no stop condition
          holds, and the obstacle is closer
          than 1.5 m or the motor temperature
          is above 75 degrees C."},
      {"id": "stop", "description": "Stop
          immediately: the obstacle is closer
          than 0.5 m, or the motor temperature
           is above 90 degrees C."}
    ]
  }
}
```

IV. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections IV-A to IV-H below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m^2” or “webers per square meter”, not “webers/m^2”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm^3”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

D. LaTeX-Specific Advice

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

BIBTEX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BIBTEX to produce a bibliography you must send the .bib files.

LATEX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LATEX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [7].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 2", even at the beginning of a sentence.

TABLE II
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

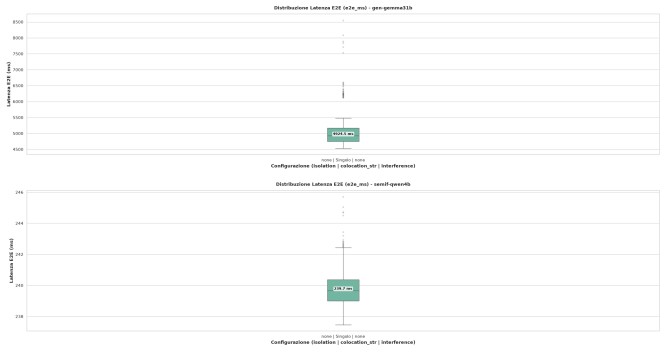


Fig. 2. Example of a figure caption.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks . . .”. Instead, try “R. B. G. thanks. . .”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first . . .”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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